

Victor Zhou

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Final-year engineering student at Polytech Sorbonne (Applied Mathematics & Computer Science), I am looking for a **6-month Data Science and AI/Machine Learning internship starting in March 2027**. Passionate about AI and its real-world applications, I have gained experience in generative AI (RAG, LLM), Data Science, Full-Stack development and DevOps through my experiences at BPCE-IT, INSERM and LIP6 (CNRS), which I wish to deepen in a stimulating environment.

Education

Polytech Sorbonne, Sorbonne Université (Paris, FRANCE) & **Kyoto University** (Kyoto, JAPAN) 2024 – 2027

- Master of Engineering in Applied Mathematics and Computer Science: Statistics, Optimization, Probability, Data Analysis, Databases, High-Performance Computing, Advanced Algorithms, Security, Numerical Analysis,
  International exchange semester at Kyoto University, Japan (Oct. 2026 – Feb. 2027)

Polytech Sorbonne, Sorbonne Université (Paris, FRANCE) 2022 – 2024

- Integrated preparatory program, PeiP - General track: Mathematics, Physics, Computer Science, Chemistry, Electronics

Experience

Software Engineer Intern - AI & Innovation, BPCE-IT, Paris May 2026 – Aug. 2026

- Design and deployment of a chatbot based on a multi-service RAG architecture using the `ChromaDB` vector database, dedicated to analyzing the company's internal documentation (Minimum Viable Product)
- Full-Stack development of business applications, including an internal coaching platform (slot booking): `FastAPI` REST API and `React` interfaces, covering all the roles of Product Owner, Full-Stack Dev, UX/UI and DevOps
- Implementation of automated CI/CD pipelines (`Jenkins`, `XLD/XLR`) ensuring continuous integration, testing and delivery. Deployment, management and orchestration of containers on enterprise `OpenShift` (`Kubernetes`) clusters

Industrial project, Software Engineer & AI, INSERM - Polytech Sorbonne, Paris Sept. 2025 – May 2026

- Development in `Python` of a software solution automating PDF text extraction, analysis and bibliographic consistency checking in INSERM's expert reports
- Implementation of a complete RAG architecture: chunking of source PDFs, embedding generation with `sentence-transformers` and semantic search using cosine similarity
- Integration of LLMs with the `Mistral` API and locally with `Ollama` for data confidentiality in order to generate consistency scores and justifications, with creation of automated deployment scripts

Research Assistant Internship, LIP6 - CNRS - Sorbonne Université, Paris May 2025 – Aug. 2025

- Implementation in `Go` of a distributed algorithm fostering the emergence of hubs in P2P networks
- Automation of simulated and real-world benchmarks evaluating network stability and convergence under various constraints, including hub loss. In-depth analysis of system metrics (%CPU, CPU Time, RAM)

Projects

Data Analysis & Machine Learning, Polytech Sorbonne May 2025

- Data processing and exploratory data analysis (EDA) with `pandas` on a Spotify dataset (Kaggle)
- Implementation and evaluation of a Machine Learning model to predict a song's popularity with `scikit-learn`

Study of the Evolution of Malaria, Polytech Sorbonne March 2025 – May 2025

- Modeling of a nonlinear differential system and numerical solving in `Python` (Euler, Runge-Kutta 4)
- Impact analysis on real WHO data and formulation of strategic public health recommendations (vaccines)

Skills

Programming Languages: `Python`, C, C++, `Go`, SQL, R, RISC-V Assembly, Bash
AI & Data Science: Machine Learning, LLMs, RAG, vector databases, `sentence-transformers`, `scikit-learn`, `pandas`
Full-stack Dev, Cloud & DevOps: `FastAPI`, `React`, Docker, `OpenShift` (`Kubernetes`), `Jenkins` (CI/CD), `XLD/XLR`, Git/GitHub, GitLab, Linux, Makefile, LaTeX, Agile Methodology
Languages: French (native), English (TOEIC 940/990), Chinese (A2)

Interests

- **Student Involvement & Sport:** Member of the Sports Office (Polytech Sorbonne, 2024–2025): Event organization and teamwork; Club football (12 years): Team spirit, discipline and rigor.
- **Passions:** Technology watch (AI, Data, Economics); Chess; Sports; Travel and discovering new cultures.